WhatsApp Video Call UI Behavior

https://gemini.google.com/app/4c2c3dc7cb60514e

User prompt: exaplain the behaviour on how video call ui behaive on whatssap

Response: WhatsApp video call UI is designed to be minimal, keeping the video full-screen while placing common overlay controls at the top and bottom.

Key Layout & UI Behaviors

1. Main View & Camera Feed

- **Full-Screen Video:** The person you are talking to takes up the entire screen background.
- **Picture-in-Picture (Self-Preview):** Your own camera view appears as a small floating window in one of the corners.
    - *Interaction:* You can tap and drag your self-preview box to any corner of the screen so it doesn't block the other person.
    - *Double Tap:* Double-tapping your self-preview usually flips or enlarges the view.

2. Control Bar Overlay

Tapping anywhere on the screen toggles the visible control overlay:

- **Bottom Action Bar:**
    - **Mute Microphone (Microphone Icon):** Toggles audio on or off.
    - **Camera Switch (Flip Icon):** Switches between the front-facing and rear-facing cameras.
    - **Disable Video (Video Cam Icon):** Turns off your camera, switching your stream to your profile picture while keeping audio live.
    - **End Call (Red Phone Icon):** Immediately disconnects the call.
- **Top Bar:**
    - **Back/Minimize Arrow:** Minimizes the video call into a small floating window so you can navigate WhatsApp chats or use other apps while staying on the call.
    - **Add Participant (Person + Icon):** Allows you to invite additional contacts to turn the call into a group video call.
    - **Encryption Notice:** Shows end-to-end encryption status.

3. Screen Sharing & In-Call Extras

- **Screen Sharing Icon:** Located in the bottom control tray on supported devices; allows you to share your screen content with the other call members.
- **Effects & Filters:** On supported devices, an effects icon lets you apply background blur or low-light fixes dynamically.

4. Background & Multitasking Behavior

- **Minimizing the App:** If you leave WhatsApp during a call, the video transitions into System Picture-in-Picture mode (a small floating window over your phone's home screen or other apps).
- **Network Status Alerts:** If your connection drops or weakens, WhatsApp overlays a banner at the top displaying messages like *"Reconnecting..."* or *"Poor connection"*, occasionally pausing video feeds to conserve audio stability.

User prompt: i just need the picture in picture behavior

Response: WhatsApp uses Picture-in-Picture (PiP) to let you multitask outside the app during an active video call.

How the Behavior Works

- **Activation:** When you are on a video call and swipe up from the bottom of your screen (or press your device's home button) to leave WhatsApp, the app automatically shrinks the live video feed into a compact floating window.
- **System Overlay:** The floating video tile hovers over your phone's home screen, web browser, or other apps so you can keep seeing the other person while doing other things.
- **Movement:** You can drag and position the floating PiP window anywhere along the edges of your screen so it doesn't block whatever you are trying to look at.
- **Quick Controls:** Tapping the floating window usually brings up minor overlay controls, such as a button to expand back to full-screen WhatsApp, mute your mic, or hang up.
- **Returning:** Tapping the window once and selecting the expand icon (or tapping the window directly on some systems) brings WhatsApp back to the foreground in full-screen.

Sources

1. WhatsApp rolls out picture-in-picture video call feature for these users

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User prompt: can this be added to my custom web app

Response: Yes, you can add this exact behavior to your custom web app! Modern web browsers have a built-in **Picture-in-Picture API** that allows any HTML5 `<video>` element to pop out into a floating window over the operating system or other browser tabs.

Here is how it works under the hood for web apps:

1. **The Native API:** Browsers provide a standard method on video elements called `requestPictureInPicture()`.
2. **Triggering It:** You can attach this method to a custom button in your UI so that when a user clicks it (or when they leave the tab/window, using the Page Visibility API), the video pops out into a floating PiP window.
3. **Browser Support:** It works natively in Chrome, Edge, Safari, and Firefox on both desktop and mobile devices that support system PiP overlays.

User prompt: is there ready made copunet that dose all this so i dont build from sctrach

Response: Yes, there are several open-source components and WebRTC SDKs that handle Picture-in-Picture (PiP) and video calling out of the box so you don't have to build it from scratch.

1. Ready-Made Video UI SDKs (Fastest & Most Complete)

If you are building a full video calling app, full-stack video WebRTC providers include built-in PiP mode, UI layouts, and grid systems:

- **VideoSDK (`@videosdk.live/react-sdk`):** Has a dedicated React / React Native hook for PiP mode. It handles background state, switching between full call view and PiP, and stream swapping automatically.
- **Agora Web SDK (`agora-rtc-react`):** Agora provides pre-built UI components and code samples for floating video overlays and PiP toggles.
- **Daily.co (`@daily-co/daily-react`):** Offers ready-to-use call UI components with built-in Picture-in-Picture support for video streams.

2. Standalone UI & Controls Libraries

If you already have your video stream logic (like WebRTC or LiveKit) and just need ready-made UI components:

- **Media Chrome (`media-chrome` by Mux):** A set of customizable Web Components and React wrappers (`<media-pip-button>`). It integrates directly with standard video elements to provide native PiP toggles without extra custom JavaScript.
- **`react-use-pip`:** A lightweight React hook that adds one-click PiP support, browser compatibility checks, and active PiP state tracking to standard HTML `<video>` elements.

3. Native Web APIs (Zero Dependencies)

If you want to keep your bundle size small without extra packages, modern browsers expose the **Document Picture-in-Picture API** and standard `<video>.requestPictureInPicture()`. You can attach a click handler directly to your video element:

```
// Toggle PiP on any video element
async function togglePiP(videoElement) {
  if (document.pictureInPictureElement) {
    await document.exitPictureInPicture();
  } else {
    await videoElement.requestPictureInPicture();
  }
}
```

Sources

1. Picture-in-Picture Mode - React Native - Video SDK
2. Building Picture-in-Picture (PiP) Mode in React Native with Expo

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